

SANDEEP BARIGE *Generative AI Engineer | Python Backend Engineer*

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PROFILE

Generative AI and Python engineer with hands-on experience building FastAPI services, RAG workflows, LLM evaluation pipelines, and database-backed applications. Delivered a production-oriented backend for a fitness client and developed AI systems using LangGraph, semantic search, FAISS, Pydantic, PostgreSQL, Firestore, and Docker.

SKILLS

GenAI & LLMs: LangChain, LangGraph, OpenAI API, Gemini API, RAG, FAISS, Embeddings, LLM-as-Judge, Hugging Face
ML: Scikit-learn, TensorFlow, XGBoost, NLP, Pandas, NumPy
Backend: Python, FastAPI, Django, Flask, REST APIs, JWT Auth, Pydantic, Redis
Databases: PostgreSQL, MySQL, Firebase Firestore
Cloud/DevOps: Docker, GitHub Actions, AWS (EC2, S3), GCP, Linux

PROFESSIONAL EXPERIENCE

SYGMS Fitness Studio (Contract, Remote) Jan 2026 – Mar 2026
Freelance Backend Developer

- Built a modular FastAPI backend on Firebase Firestore covering member onboarding, memberships, renewals, payments, and dashboard statistics
- Implemented PIN-based auth (hashed login, OTP verification, reset) and payment idempotency that prevented duplicate revenue counting
- Containerized the deployment with Docker and added a protected cron endpoint for membership-expiry and renewal reminders

National Small Industries Corporation (NSIC), India May 2022 – Aug 2023
AI & ML Intern

- Built Python ETL pipelines using Pandas and NumPy to process structured datasets and reduce manual data correction effort by approximately 25%.
- Developed validation scripts to detect missing values, inconsistent records, schema mismatches, and data-quality issues before reporting and model workflows.
- Built an NLP spam classification workflow using FP-Growth and Naive Bayes for text preprocessing, feature extraction, classification, and prediction.

PROJECTS

AgentOps Monitor — AI Agent Evaluation and Observability Platform Jun 2026 – Present
FastAPI, LangGraph, OpenAI, PostgreSQL, Redis, FAISS, Docker, React

- Built the core FastAPI backend with PostgreSQL persistence and instrumentation capturing agent runs, LLM calls, tool executions, latency, token usage, and errors
- Designed LLM-as-judge evaluation workflows for relevance, faithfulness, hallucination risk, and task completion
- Currently implementing RAG-based verification with FAISS to compare generated responses against retrieved context and flag unsupported claims

AI Powered Personalized Learning Platform Jan 2025 – Dec 2025
Django, FastAPI, OpenAI, Gemini, PostgreSQL, FAISS, Sentence Transformers

- Architected a hybrid Django and FastAPI platform separating core learning workflows from AI-powered services.
- Developed an LLM-powered course-generation pipeline and a RAG-based tutoring workflow grounded in retrieved course context.
- Added structured-output instructions and Pydantic validation to improve the reliability of generated JSON.
- Developed automated grading using semantic similarity, task-completion checks, code-quality checks, and optional Docker-sandboxed execution.

E-Library Search Engine for Books & Articles Jan 2025 – May 2025
Python, Flask, MySQL, Whoosh, NLTK, HTML/CSS

- Built a full-text academic search engine with document indexing, query preprocessing, Boolean search, relevance scoring, user authentication, search history, and favorites.
- Implemented text preprocessing workflows using NLTK and backend APIs using Flask, MySQL, and SQLAlchemy.

Customer Churn Prediction System May 2024 – Jul 2024
Python, Pandas, NumPy, Scikit-learn, Random Forest, Gradient Boosting,

- Logistic Regression, Data Preprocessing, Model Evaluation.
- Built an end-to-end supervised machine learning pipeline to predict customer churn using demographic, account, and transactional data.
 - Performed data cleaning, missing-value handling, encoding, scaling, feature engineering, train-test splitting, and model evaluation.
 - Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix analysis, and error analysis.

EDUCATION

Master of Science in Computer Science Jan 2024 – Dec 2025 | USA
Lawrence Technological University

Bachelor of Technology in Computer Science and Engineering Jun 2019 – Jun 2023 | India
GITAM Deemed University